Middle-Rung Law predictions — preregistration text

OSF Preregistration — “Middle-Rung Law” predictions

Title: Predictions of a three-rung frequency ladder linking the Hubble rate, the Planck frequency and an “ether” frequency **Date of preregistration:** 2026-11-05 (predictions fixed in a private git repository on that date; this text makes them public)

1. Hypothesis

The Planck frequency is the geometric mean of the cosmic expansion rate H and an ether frequency ν_e: $\nu_e \cdot H = \nu_P^2$ at every epoch, equivalently $\hbar G = c^5/(4\pi^2 \nu_e H)$. ν_e equals the de Broglie angular frequency of the Hubble sphere treated as one object (Kepler’s third law with $r = c/H$, $T = 2\pi/H$, combined with $Mc^2 = h\nu$; normalization relation written by Zhang Xiangqian, 2026). The expansion follows standard early-universe physics and switches to coasting, $H \propto (1+z)$, near $z \approx 1.6$–1.7.

2. Predictions (fixed 2026-11-05)

ID	Quantity	Predicted value	Deciding data	Falsified if
P1	H_0	61.8–63.5 km s^{-1} Mpc^{-1}	gravitational-wave standard sirens	measured value outside range at 3σ (e.g. 73)
P2	age of universe	13.5–13.9 Gyr	stellar ages + independent cosmology	reliably < 13.4 or > 14.0 Gyr
P3	expansion history	coasting below $z \approx 1.6$; at $z = 2.33$, H about 3% below ΛCDM	DESI final, Euclid, Roman BAO	z 1.5–2.5 consistent with ΛCDM at < 1% precision, no transition
P4	dark matter	compact extreme objects, 10^{14}–10^{20} kg	microlensing, femtolensing, sub-solar GW searches	window fully excluded
P5	MOND-like acceleration scale	$a_0 = (0.96–0.98)\times10^{-10}$ m s^{-2} today, $\propto H(z)$: ×2 at $z = 1$, ×3 at $z = 2$	high-z rotation curves	no redshift evolution

Derived: $N = \nu_P/H = (1.44–1.47)\times10^{60}$; $\nu_e = (4.2–4.4)\times10^{102}$ Hz; mid-rung frequency $\sqrt{(H \nu_P)} = 2.43–2.46$ THz. Ether-flow (gravitational potential) changes add in quadrature; under current axioms a dual-optical-clock comparison next to any non-mass device should read zero.

3. Disclosure of data already examined

Data for P3 and P5 already existed. After fixing the predictions we compared them on 2026-11-05: - P5: Ciocan et al. 2026 (Astronomy and Astrophysics 709, L16) find a_0 rising with z, 2.38×10^{-10} at z ≈ 1. Direction agrees; magnitude about 19% above our prediction. Earlier analyses (Milgrom 2017; Limbach et al. 2008) disfavour strong evolution. - P3: DESI DR2 BAO (13 points). Best $\chi^2 = 27.7$ for our model vs 11.5 for ΛCDM with free Ω_m and r_d. This is unfavourable to our model, with the largest residual at z = 0.934. - P1: current GW-only constraints (±15–20) do not discriminate. We will record every future comparison, favourable or not, without changing the predictions above.

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